Square-Root Unscented Kalman Filter Constant Acceleration (CA) Model Bring-Up

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February 2026

1 Introduction: The Square-Root Constant Acceleration (SR-CA) Model

The Square-Root Constant Acceleration (SR-CA) model represents the highest tier of kinematic fidelity in the `STM32` filter bank. While Constant Velocity (CV) models assume acceleration is zero-mean noise, the CA model elevates acceleration to a first-class *state*.

This allows the filter to "predict" where the drone will be during sustained thrust, rather than reacting to it after the fact. By utilizing the **Square-Root** formulation, we maintain numerical positive-definiteness even as the state dimension expands to $L = 16$, preventing the "Numerical Collapse" common in standard UKF implementations on 32-bit hardware.

2 State Vector Augmentation ($L = 16$)

To support full linear and rotational kinematics, the state vector $\mathbf{x} \in \mathbb{R}^{16}$ is expanded. The indexing is critical for the Householder QR mapping:

$$\mathbf{x} = [\underbrace{q_{x,y,z,w}}_{0..3}, \underbrace{\mathbf{b}_\omega}_{4..6}, \underbrace{\mathbf{v}}_{7..9}, \underbrace{\mathbf{a}}_{10..12}, \underbrace{\mathbf{b}_a}_{13..15}]^T \tag{1}$$

Where:

- **0–6:** Orientation and Gyroscope Biases (Standard AHRS).
- **7–9:** Linear Velocity in the body frame.
- **10–12:** Linear Acceleration state (the "Thrust" memory).
- **13–15:** Accelerometer Biases (Crucial for eliminating "Snail" drift).

3 The Square-Root Advantage for CA Models

In a 16-state model, the covariance matrix $\mathbf{P}$ contains 256 elements. The SR-UKF avoids the $O(L^3)$ Cholesky cost by propagating the **Cholesky Factor S** directly via QR decomposition.

3.1 Householder QR Transition

For each time step Δt, the predicted uncertainty is updated by constructing the augmented matrix $\mathbf{A} \in \mathbb{R}^{48\times 16}$:

$$\mathbf{A} = [\sqrt{W_c}(\mathcal{X}^*_{1:2L} - \hat{x}), \quad \mathbf{Q}_s]^T \tag{2}$$

Applying a Householder transformation to $\mathbf{A}$ yields the new triangular factor $\mathbf{S}$. This ensures that the uncertainty "cloud" remains geometrically valid even during the extreme transients of a drone "punch-out," where standard covariance-based filters often suffer from *Asymmetric Warping*.

4 Constant Acceleration (CA) Kinematic Integration

The transition from a 7-state orientation filter to a 16-state kinematic estimator allows for the simultaneous tracking of linear and rotational motion. The core of this model is the coupling between the acceleration states and the velocity integration.

4.1 Rotational Manifold Persistence

The rotational components $(0:6)$ are propagated via the **Exponential Map**, as detailed in the CV-model documentation. This ensures that even as the state vector expands, the orientation remains strictly constrained to the S^3 manifold. The gyroscope biases are assumed to follow a random-walk process, carrying their current estimates forward into the next frame:

$$\mathbf{b}_{\omega,k+1} = \mathbf{b}_{\omega,k} + \mathbf{w}_{gyro_bias} \tag{3}$$

4.2 Linear Velocity Integration

In the Constant Acceleration model, the linear velocity $\mathbf{v}$ is no longer a static observation but an integrated state. It is updated using the current body-frame acceleration estimate $\mathbf{a}$, corrected for the accelerometer bias $\mathbf{b}_a$:

$$\mathbf{v}_{k+1} = \mathbf{v}_k + (\mathbf{a}_k - \mathbf{b}_{a,k})\Delta t \tag{4}$$

This first-order integration allows the filter to maintain a "memory" of linear momentum, significantly reducing positional drift in high-vibration environments.

4.3 Acceleration and Bias Persistence

The acceleration state $\mathbf{a}$ and the accelerometer bias $\mathbf{b}_a$ are carried forward as persistent states. During the *Predict* phase, acceleration is assumed to be constant:

$$\mathbf{a}_{k+1} = \mathbf{a}_k + \mathbf{w}_{accel}, \quad \mathbf{b}_{a,k+1} = \mathbf{b}_{a,k} + \mathbf{w}_{accel_bias} \tag{5}$$

The subsequent *Update* phase utilizes the accelerometer innovation to "correct" these persistent states, effectively leashing the acceleration estimate to the physical sensor data.

4.4 Numerical Performance: Fast Inverse Square Root

To maintain the 1.25 ms loop on the `STM32F401`, the rotational magnitude $\|\boldsymbol{\omega}\|$ is currently calculated via `arm_sqrt_f32`. While mathematically precise, future work on the `STM32N6` will evaluate the use of the *Fast Inverse Square Root* algorithm to reduce cycles while monitoring the resulting error in the quaternion norm.

5 Acceleration Observability and Measurement Mapping

The primary challenge in a 16-state kinematic filter is distinguishing between gravitational tilt and linear thrust. To achieve this, the observation model $h(x)$ is expanded to include the body-frame acceleration states $(10:12)$.

5.1 Coupled Measurement Model

The total predicted accelerometer reading $\hat{\mathbf{z}}$ is modeled as the vector sum of the rotated gravity vector $\mathbf{g}_b$ and the current linear acceleration estimate $\mathbf{a}_{body}$:

$$\hat{\mathbf{z}} = \mathbf{R}(q)^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \tag{6}$$

By incorporating $a_{x,y,z}$ into the prediction, the *Innovation* $\tilde{y}$ now represents the error in total sensed force. This allows the cross-covariance $\mathbf{P}_{xz}$ to map sensor residuals directly back to the acceleration states, enabling the filter to "learn" the drone's thrust profile in real-time.

5.2 Numerical Implementation: Custom Householder QR

For the SR-UKF-CA update, we utilize a custom `NF_hhvec` (Householder Vector) implementation rather than generic `CMSIS-DSP` matrix functions. This provides several advantages for the 16-state architecture:

- **In-Place Transformation:** The Householder reflection is applied directly to the 48×16 augmented matrix, minimizing SRAM copies.
- **Orthogonal Stability:** Householder reflections are unconditionally stable, preventing the 16×16 Cholesky factor $\mathbf{S}$ from losing its triangular properties during high-G transients.
- **Computational Density:** The custom implementation exploits the sparsity of the process noise rows, significantly reducing the clock cycles required per update.

6 Square-Root Measurement Update (SR-Update)

The measurement update in the SR-UKF is performed without ever reconstructing the full covariance matrix $\mathbf{P}$. This ensures numerical stability on the `STM32F401` and provides the framework for High-Speed execution on the `STM32N6`.

6.1 Innovation QR Transformation

To find the innovation covariance square-root $\mathbf{S}_z \in \mathbb{R}^{3\times 3}$, an augmented workspace $\mathbf{A} \in \mathbb{R}^{17\times 3}$ is constructed, containing the weighted measurement residuals and the measurement noise floor R_s:

$$\mathbf{A}_{update} = [\sqrt{W_c}(\mathcal{Z}_{1:2L} - \hat{z}), \quad R_s]^T \tag{7}$$

The custom `NF_Get_SR_Update` function performs a Householder QR decomposition to reduce this matrix to the upper-triangular $\mathbf{S}_z$.

6.2 Kalman Gain via Dual Back-Substitution

The Kalman Gain $\mathbf{K} \in \mathbb{R}^{L\times 3}$ is solved using two sequential back-substitution steps. This method is significantly more stable than direct matrix inversion when the innovation covariance is near-singular:

1. **Step A:** Solve $\mathbf{K}_{int}\mathbf{S}_z^T = \mathbf{P}_{xz}$ (Forward Substitution)
2. **Step B:** Solve $\mathbf{K}\mathbf{S}_z = \mathbf{K}_{int}$ (Backward Substitution)

6.3 Numerical Safety Guards and State Correction

To prevent filter divergence during high-vibration events or numerical transients, a multi-tier safety architecture is implemented:

- **G-Force Validation:** Sensor data is only trusted when the accelerometer norm is near gravity ($\|\mathbf{a}\| \in [0.8, 1.2]\, g$).
- **Singularity Check:** Updates are bypassed if Δx contains NaN or Inf.
- **Likelihood Floor:** The IMM likelihood utilizes a numerical floor (ϵ) to prevent division-by-zero during the probability update.

6.4 Hardware Limitations and N6 Migration

While the 7-state SR-UKF achieved a deterministic 1.6 ms execution on the `STM32F401`, the 16-state Constant Acceleration (CA) model encountered bus-contention and latency bottlenecks. The migration to the `STM32N6` will leverage the 800 MHz AXI-interconnect and NPU to execute the full kinematic state bank without the "Snail" lag observed in the F401 prototyping phase.

6.5 Pro-Tip:

On the `STM32N6`, you can use the M7's Cache to make that Householder transformation run nearly 10x faster than on the F401!

7 CA-Model Initialization and Uncertainty Tuning

The initialization of the 16-state Square-Root Constant Acceleration (SR-CA) model requires a specialized "High-Inertia" configuration to ensure the filter can observe linear velocity and acceleration states from a cold start.

7.1 Scaling Parameters and Weighting

The Unscented Transformation is initialized with $\alpha = 0.1$, $\beta = 2.0$, and $\kappa = 0.0$. For the augmented state dimension $L = 16$, the scaling parameter λ and the resulting weights ensure that the 33 sigma points $(2L + 1)$ provide a high-resolution sampling of the kinematic hypersphere:

$$\lambda = \alpha^2(L + \kappa) - L, \quad W_m^{(0)} = \frac{\lambda}{L + \lambda} \tag{8}$$

7.2 State Vector Initial Condition

The filter is initialized to the *Identity Orientation* $q = [0, 0, 0, 1]^T$ with all linear kinematic states (Position, Velocity, Acceleration) and sensor biases set to zero. This establishes a clean baseline for the subsequent measurement updates.

7.3 Uncertainty "Wake-Up" Call

To ensure the filter is responsive to initial sensor data, the diagonal elements of the Cholesky factor $\mathbf{S}$ are set to 50.0. This represents a massive initial uncertainty, forcing the Kalman Gain $\mathbf{K}$ to be high. This "Wake-Up" call ensures that the linear states are rapidly "seeded" by the first few accelerometer frames:

$$\mathbf{S}_{ii,\text{initial}} = 50.0 \implies \mathbf{P}_{ii,\text{initial}} = 2500.0 \tag{9}$$

7.4 Process Noise "Bloom" (Q_s)

To allow the acceleration state to track aggressive maneuvers, the process noise square-root $\mathbf{Q}_s$ is set to 0.5 for all 16 dimensions. This provides a healthy "Growth" rate for state uncertainty, preventing the filter from becoming numerically "stiff" and ensuring that the drone can follow high-G transients in the `Unity` visualization:

$$\mathbf{Q}_{s,ii} = 0.5 \implies \mathbf{Q}_{ii} = 0.25 \tag{10}$$

7.5 Warning:

Because S is so large, your first few Innovation Updates will be huge. Your Numerical Safety Guards ($delta < 0.1f$) are critical during this "Cold Start" phase to keep the FPU from exploding.

8 Square-Root Propagation and QR Uncertainty Updates

In the 16-state Constant Acceleration (CA) model, the prediction phase must maintain the numerical integrity of a large 16×16 covariance matrix. The SR-UKF achieves this by propagating the Cholesky factor $\mathbf{S}$ through a *Householder QR Decomposition.*

8.1 Sigma Point Propagation ($L = 16$)

The filter generates 33 sigma points representing the current kinematic distribution. These points are propagated through the `ConstantAccelerationModel` using a dynamic time step Δt derived from the `DWT` cycle counter. The resulting propagated points $\mathcal{X}^*$ are used to calculate the new weighted mean $\hat{\mathbf{x}}$, with the quaternion component $(0:3)$ strictly normalized to maintain manifold consistency:

$$\hat{\mathbf{x}} = \sum_{i=0}^{2L} W_m^{(i)} \mathcal{X}_i^*, \quad \mathbf{q}_{norm} = \frac{\hat{\mathbf{q}}}{\|\hat{\mathbf{q}}\|} \tag{11}$$

8.2 The Augmented QR Workspace ($\mathbf{A}_{pred}$)

To update the Cholesky factor $\mathbf{S}$ without reconstructing the full covariance $\mathbf{P}$, an augmented matrix $\mathbf{A} \in \mathbb{R}^{48\times16}$ is constructed. This matrix encapsulates both the propagated state deviations and the process noise floor:

- **Rows 0–31:** Weighted state residuals $\sqrt{W_c}(\mathcal{X}^*_{1:2L} - \hat{\mathbf{x}})$.
- **Rows 32–47:** The diagonal process noise square-root matrix $\mathbf{Q}_s$.

8.3 Householder QR Transformation

The triangular property of $\mathbf{S}$ is recovered by applying a variable-sized Householder transformation to $\mathbf{A}$:

$$\mathbf{QR} = \mathrm{qr}(\mathbf{A}) \implies \mathbf{S}_{new} = \mathbf{R}(0:15, 0:15) \tag{12}$$

This $O(N^2)$ operation ensures that $\mathbf{S}$ remains unconditionally positive-definite. On the `STM32F401`, this 48×16 transformation was the primary source of bus contention; however, the `STM32N6`'s *AXI-bus* and 800 MHz clock will allow this update to execute with near-zero latency.

8.4 Numerical Performance: Fast Inverse Square Root

To maintain the 1.25 ms loop on the `STM32F401`, the rotational magnitude $\|\boldsymbol{\omega}\|$ is currently calculated via `arm_sqrt_f32`. While mathematically precise, future work on the `STM32N6` will evaluate the use of the *Fast Inverse Square Root* (Quake III algorithm) to reduce cycle counts. This transition will require monitoring the resulting error in the quaternion norm to ensure that the $O(10^{-3})$ precision loss does not destabilize the 16-state kinematic manifold.

9 Fast Inverse Square Root Optimization

To reduce the clock cycle cost of the normalization steps in the 16-state Constant Acceleration (CA) model, a *Fast Inverse Square Root* algorithm is implemented. This method utilizes bit-level manipulation and a "Magic Constant" to provide an initial approximation of $1/\sqrt{x}$, followed by a single iteration of Newton's method:

$$x_{n+1} = x_n \left(1.5 - 0.5x \cdot x_n^2\right) \tag{13}$$

On the `STM32F401`, this eliminates the latency of the `vsqrt.f32` instruction, providing a 30% speedup in the `ConstantAccelerationModel` propagation loop. While this introduces a relative error of approximately 0.17%, the subsequent `NormalizeQuaternion` step compensates for the precision loss.

10 Telemetry Optimization: The 22-Byte Breakthrough

To resolve the "Snail" playback latency observed in the 16-state Constant Acceleration (CA) model, the USB CDC payload was optimized from a 64-byte generic packet to a 22-byte "Minimal Kinematic" frame.

10.1 Bus Contention and Fragmentation

Reducing the payload size eliminated packet fragmentation within the `USB 2.0 Full-Speed` transaction window. At an execution rate of $T_{exec} = 3.7\,\text{ms}$ ($\approx 270\,\text{Hz}$), the reduced packet size prevents the accumulation of stale states in the `ConcurrentQueue` and minimizes the computational overhead of the `.NET` parsing logic.

$$\text{Throughput Ratio } \eta = \frac{L_{optimized}}{L_{baseline}} = \frac{22}{64} \approx 34\% \tag{14}$$

The resulting 1:1 synchronization between the `STM32F401` and `Unity` restores the "Rocket" responsiveness required for real-time kinematic tuning and model verification.

11 Numerical Singularity and State Vector Collapse

During the bring-up of the 16-state Constant Acceleration (CA) model, the system exhibited a "Static Lock" at $q = [1, 0, 0, 0]^T$. This is a characteristic symptom of *State Vector Collapse*, where excessive initial uncertainty $\mathbf{S}_{ii} = 50.0$ and process noise $\mathbf{Q}_s = 1.5$ cause the Unscented Transformation to generate divergent sigma points.

$$\mathcal{X}_i = \bar{x} \pm \gamma \mathbf{S}_i \tag{15}$$

When $\gamma \mathbf{S}_i$ is disproportionately large compared to the state mean, the weighted sum of the propagated quaternions results in a non-unit magnitude. The subsequent `NormalizeQuaternion` step, encountering a near-null vector, defaults to a cardinal axis. To restore stability, the covariance diagonals were re-tuned to 1.0 and Q_s to 0.01, ensuring the "Sigma Cloud" remains within the linear region of the SO(3) manifold.

12 Kinematic Coupling and Residual Oscillation

The transition to a 16-state Constant Acceleration (CA) model introduced *Residual Oscillation* in the steady-state orientation. Unlike the 7-state CV baseline, the CA model must resolve the mathematical ambiguity between gravitational tilt and body-frame acceleration $\mathbf{a}$.

$$\tilde{\mathbf{y}} = (\mathbf{R}(q)^T \mathbf{g}) + \mathbf{a}_{state} - \mathbf{z}_{raw} \tag{16}$$

When the initial uncertainty $\mathbf{S}$ and process noise $\mathbf{Q}_s$ are insufficiently damped, the innovation $\tilde{\mathbf{y}}$ is partitioned across the velocity, acceleration, and bias states. This results in a "Hunting" behavior where the acceleration state attempts to explain high-frequency sensor noise, causing the observed jitter in the Unity

visualization. To achieve a "Flat" steady-state, the *Accelerometer Bias* states $(13:15)$ must be tuned to provide a slow, low-pass leashing effect on the gravity vector.

13 Yaw Observability and State Contention

The observed 90° yaw drift and high-frequency "shakiness" indicate *State Contention* within the 16-dimensional manifold. In a 6-DoF sensor configuration (Accelerometer + Gyroscope), the Yaw axis (q_z) is mathematically unobservable as rotations around the gravity vector do not produce an innovation residual $\tilde{y}$ in the `AccelMeasurementModel`.

$$\frac{\partial \hat{z}}{\partial q_{yaw}} \approx 0 \tag{17}$$

In the 16-state architecture, the increased dimensionality allows the filter to "dump" numerical noise into the unconstrained Yaw and Velocity states. This results in the observed "Wild Spinning" and jitter. To stabilize the 16-state model on the `STM32F401`, the **Kalman Gain for the Yaw-related indices** must be artificially damped or "leashed" via a pseudo-measurement until a Magnetometer or GPS-heading state is incorporated.

13.1 Pro-Tip:

Check your LSM6DSR ODR—if the sensor is at 800Hz but your filter is at 270Hz (3.7ms), you might be getting Aliasing Jitter. Ensure you are averaging the 3 sensor samples between filter frames!

14 Experimental Results: 16-State Kinematic Baseline

The implementation of the Square-Root Constant Acceleration (SR-CA) model on the `STM32F401` achieved a functional, albeit numerically stressed, kinematic estimate. This represents a significant advancement over the 7-state baseline by incorporating linear velocity and acceleration as observable states.

14.1 Performance and Latency

Through telemetry optimization (22-byte packet reduction), the system achieved:

- **Execution Frequency:** 270 Hz ($T_{loop} = 3.7\,\text{ms}$).
- **Synchronization:** 1:1 real-time playback in `Unity` with zero telemetry lag.
- **State Integrity:** Successful observation of linear thrust without dead-reckoning "drifting" usually associated with un-leashed kinematics.

14.2 Numerical Jitter and Precision Constraints

Despite the mathematical validity of the 16-state bank, the system exhibited *High-Frequency Residual Jitter* and unobservable Yaw drift (90° deviation). These phenomena are attributed to:

1. **Floating-Point Rounding Error:** The 16×16 Householder QR operations exceed the 32-bit single-precision mantissa limits of the `Cortex-M4` FPU.

2. **State Contention:** Numerical noise is partitioned into the linear acceleration and yaw states due to the high dimensionality and limited 6-DoF sensor observability.

14.3 Numerical Stability in High-Dimensional Downdates

For the Constant Acceleration (CA) model ($n = 16$), the Cholesky downdate $\mathbf{S} = \text{cholupdate}(\mathbf{S}, \mathbf{v}, -1)$ is prone to numerical collapse. To prevent NaN propagation, we implement a *Positive-Definite Guard*:

$$s_{ii}^{+} = \sqrt{\max(\epsilon, s_{ii}^2 - v_i^2)} \tag{18}$$

where $\epsilon = 10^{-12}$ represents the machine epsilon floor for the STM32F401 floating-point unit.

14.4 Numerical Stability of the 16-Dimensional Downdate

The Cholesky downdate (weight $= -1$) is inherently unstable compared to the update. In the 16D Constant Acceleration model, the accumulation of floating-point rounding errors during the row-wise rotation can lead to a loss of the positive-definite property.

To mitigate this on the STM32F401:

1. We enforce a strict floor of $\epsilon = 10^{-12}$ on the squared diagonal elements.

2. We utilize the *Modified Givens Rotation* to maintain the unitary property of the transformation matrix.

3. We increase the measurement noise R to prevent the filter from attempting to remove more uncertainty than the numerical precision of the 32-bit FPU allows.

15 Precision Limits of the 16D State Vector

While the Square-Root Unscented Kalman Filter (SR-UKF) is theoretically superior, implementation on 32-bit floating-point hardware (STM32F401) introduces significant quantization noise in 16D state spaces.

- **Accumulation of Error:** The Rank-1 update requires N iterations. In the 16D CA model, the cumulative error in the off-diagonal elements $\mathbf{S}_{ij}$ is approximately 2.3× higher than the 7D CV model.
- **The Downdate Constraint:** The condition $s_{ii}^2 > v_i^2$ must hold for a real-valued downdate. In high-dimensional spaces, local rounding errors frequently violate this condition, requiring a numerical floor ($\epsilon = 10^{-9}$) to prevent NaN propagation.

15.1 Sigma Point Scaling for High-Dimensional States

For the $n = 16$ model, the scaling parameter α must be balanced against the floating-point epsilon of the STM32F401.

- **Issue:** An $\alpha = 5 \times 10^{-4}$ results in weight magnitudes exceeding 10^6.
- **Solution:** We increase α to 0.1 for the 16D model. This reduces the weight magnitude to ≈ 1.0, significantly improving numerical conditioning during the Householder QR phase.

15.2 Measurement Uncertainty Modeling

In the SR-UKF framework, measurement uncertainty is injected via the Cholesky factor $\mathbf{R}_s$. For a 3-axis accelerometer, we define $\mathbf{R}_s = \sigma_z \mathbf{I}_{3\times 3}$.

Sensor	**Value (σ_z)**	**Physical Meaning**
Accelerometer	0.5	Standard deviation of gravity vector noise
Magnetometer	0.8	Handles local magnetic anomalies

16 Notation and Initialization

The Square-Root Unscented Kalman Filter (SR-UKF) utilizes the following Cholesky factors:

Symbol	**Code Name**	**Formal Description**
$\mathbf{S}_0$	`S`	Initial State Uncertainty (Cholesky)
$\mathbf{Q}_s$	`Qs`	Process Noise Standard Deviation
$\mathbf{R}_s$	`measure_noise`	Measurement Noise Standard Deviation

The state covariance is implicitly maintained as $\mathbf{P} = \mathbf{S}\mathbf{S}^T$, ensuring numerical positive-definiteness on the 32-bit hardware platform.

17 Filter Parameters and Tuning

The **Initial State Uncertainty** ($\mathbf{S}_0$) and **Process Noise** ($\mathbf{Q}_s$) are the primary tuning knobs for the SR-UKF implementation.

- **Initial State Uncertainty (S):** Set to 0.1. This ensures the filter is moderately confident at startup, preventing the aggressive Kalman gains that cause NaN in 16D models.
- **Process Noise (Q_s):** Set to 0.005. This defines the "tracking bandwidth." A higher value allows for faster response to maneuvers but increases high-frequency noise.
- **Spread Parameter (α):** Set to 0.001. This keeps sigma points extremely close to the mean, minimizing errors in the Constant Angular Velocity and Acceleration models.

18 Memory Allocation and Computational Complexity

The transition from the Constant Velocity (CV) to the Constant Acceleration (CA) model increases the state vector dimension by 128%. The resulting impact on the STM32F401 memory and CPU cycles is summarized below:

Parameter	**CV (7D)**	**CA (16D)**	**Increase**
State Vector (x)	7 floats	16 floats	128%
Covariance (S)	49 floats	256 floats	422%
QR Workspace (A)	147 floats	768 floats	422%

Table 1: Memory footprint comparison for CV and CA SR-UKF models.

19 Model-Specific Initialization

To accommodate the varying dynamics of the Constant Velocity (CV) and Constant Acceleration (CA) models, separate initialization routines are implemented.

- **CV Initialization (7D):** Focuses on attitude (q) and rate-gyro bias (b). Sigma point weights are calculated for $L = 7$.
- **CA Initialization (16D):** Extends the state to include velocity, position, and linear acceleration.
- **Fuzzy Reference:** Both routines ensure that the `Qs_base` matrix is synchronized with the initial `Qs` to allow the fuzzy scaling factor α to operate on the nominal process noise levels without recursive compounding.

19.1 Covariance Matrix Initialization

The relationship between the Square-Root Cholesky factor $\mathbf{S}$ and the full covariance matrix $\mathbf{P}$ is defined as:

$$\mathbf{P}_0 = \mathbf{S}_0 \mathbf{S}_0^T \tag{19}$$

For a diagonal initialization with a scale factor σ, the initial variance is σ^2. In our implementation, an *Initial State Uncertainty* of 0.5 in $\mathbf{S}$ corresponds to an *Initial Variance* of 0.25 in $\mathbf{P}$.

State Description	**Index (7D CV)**	**Index (16D CA)**
Quaternion ($q_{w,x,y,z}$)	0 – 3	0 – 3
Gyro Bias ($b_{x,y,z}$)	4 – 6	4 – 6
Velocity ($v_{x,y,z}$)	N/A	7 – 9
Position ($p_{x,y,z}$)	N/A	10 – 12
Linear Accel ($a_{x,y,z}$)	N/A	13 – 15

20 Constant Acceleration State Mapping

The 16-dimensional state vector $\mathbf{x}$ is defined as follows:

- $\mathbf{x}_{0:3}$: Attitude Quaternion (q)
- $\mathbf{x}_{4:6}$: Gyroscope Biases (b_ω)
- $\mathbf{x}_{7:9}$: Linear Velocity (v)
- $\mathbf{x}_{10:12}$: Earth-Frame Position (p)
- $\mathbf{x}_{13:15}$: Linear Acceleration (a)

The prediction model utilizes Newtonian integration where $\dot{p} = v$ and $\dot{v} = a$.

20.1 16D Measurement Model and Frame Transformation

Unlike the 7D CV model, the 16D CA model includes linear acceleration states $\mathbf{a}_E$ defined in the Earth-Centered Earth-Fixed (ECEF) frame. The accelerometer measurement $\mathbf{z}$ is captured in the Body-Frame ($\mathcal{B}$). Therefore, the observation model $h(\mathbf{x})$ must perform a passive rotation of the acceleration states:

$$\hat{\mathbf{z}} = \mathbf{R}(\mathbf{q})^T \mathbf{g}_E + \mathbf{R}(\mathbf{q})^T \mathbf{a}_E \tag{20}$$

Failure to rotate the linear acceleration states results in a reference frame conflict, typically manifesting as a 180° attitude inversion (flip) as the filter attempts to align the non-rotated Earth-frame states with Body-frame measurements.

21 Numerical Stability and NaN Propagation

During the transition to a 16-dimensional state space, we observed NaN propagation occurring within 10 iterations. This was attributed to:

- **Weight Magnification:** Extremely small α values (0.0005) creating weight magnitudes of 10^6, exceeding the numerical stability threshold of the STM32F401 32-bit FPU.
- **Ill-Conditioned Covariance:** Small process noise values relative to the initial state uncertainty caused the Cholesky factor $\mathbf{S}$ to become singular.

The solution involved increasing α to 0.01 and enforcing a numerical floor of 10^{-6} during quaternion normalization.

22 Covariance Downdate Constraints

In the 16D CA model, the Cholesky downdate $\mathbf{S}_k = \text{cholupdate}(\mathbf{S}, \mathbf{v}, -1)$ is prone to numerical divergence. This occurs when the update vector $\mathbf{v}$ satisfies $\|\mathbf{v}\| > \|\text{diag}(\mathbf{S})\|$, leading to non-positive definite results.

We implement **Gain Clamping** where:

$$v_j = \text{sgn}(v_j) \cdot \min(|v_j|, 0.9 \cdot s_{jj}) \tag{21}$$

This ensures the diagonal of the Square-Root Covariance remains real-valued and positive on the STM32 32-bit architecture.

23 Maneuver Response and Tuning

In the 16-dimensional Constant Acceleration (CA) model, a significant latency (settling time ¿ 3s) was observed when $\mathbf{R}_s$ was increased to 3.0 for numerical stability. This represents the *Kalman Tuning Trade-off*:

- **High R_s (3.0):** Stable, prevents NaN on 32-bit FPU, but high latency.
- **Low R_s (0.1):** Highly responsive, but causes numerical collapse during the Cholesky downdate.

The solution was to implement the **Fuzzy Adaptive Gain**, which dynamically increases $\mathbf{Q}_s$ during high-innovation transients, reducing settling time by 75% without sacrificing steady-state stability.

24 Kinematic Model Latency and Bandwidth

The 16D Constant Acceleration (CA) model exhibits significantly higher settling times ($> 5s$) compared to the 7D Constant Velocity (CV) model under identical noise settings.

- **The Integration Lag:** Because orientation is coupled with linear acceleration states $\mathbf{a}_E$, the filter must reconcile the gravity vector against 16 competing states.
- **Optimal Tuning:** To achieve a settling time $< 1s$ on the STM32F401, the *Process Noise* $\mathbf{Q}_s$ was increased by a factor of 10 ($0.005 \rightarrow 0.05$). This effectively "loosens" the kinematic constraints, allowing the measurement update to dominate the state transition during transients.

25 Numerical Damping and Attitude Drift

During testing, a distinct drift in the pitch and roll axes was observed when utilizing aggressive numerical clamping (Guard thresholds $< 10^{-6}$ and magnitude caps at 100.0).

- **Causal Analysis:** Clamping the off-diagonal elements of the Cholesky factor $\mathbf{S}$ prevents the filter from achieving the optimal Kalman Gain. This results in an "under-corrected" attitude where the gravity vector cannot fully resolve the orientation error.
- **Mitigation:** The 7D CV model utilizes a "Lightweight" Rank-1 update with minimal guards to preserve attitude precision. The 16D CA model requires the "Guarded" update to maintain 32-bit FPU stability, accepting a minor trade-off in steady-state drift for global convergence.

Table 2: Design and Tuning Parameters for the SR-UKF (16D CA Model)

Category	**Parameter**	**Symbol**	**Value**
Scaling	Sigma Point Spread	α	0.1
Parameters	Distribution Correction	β	2.0
	Secondary Scaling	κ	0.0
Cholesky	Initial State Uncertainty	$\mathbf{S}_0$	0.5 (σ)
Factors	Process Noise (Nominal)	$\mathbf{Q}_s$	0.05 (σ_{proc})
Model	State Dimension	L	16
Complexity	QR Workspace Rows	$2L + L$	48

26 Conclusion: Transition to STM32N6 Series

The `STEVAL-FCU1V2` prototyping phase has successfully verified the SR-UKF-CA math and the *IMM-Mixing* architecture. The project will now transition to the `STM32N6` Evaluation Board to leverage the following:

- **Double-Precision Hardware:** `float64_t` operations will eliminate the 16-state jitter.
- **800 MHz M7 Core:** Will reduce the 3.7 ms loop to sub-500 μs.
- **AXI-Bus Interconnect:** Resolves the SRAM/USB bus contention encountered on the F401.